(Re)Constructing the Body: Addressing Body Concepts in the Design of Vocal Technology

COURTNEY N. REED, Digital Technologies & Creative Futures, Loughborough University London, UK

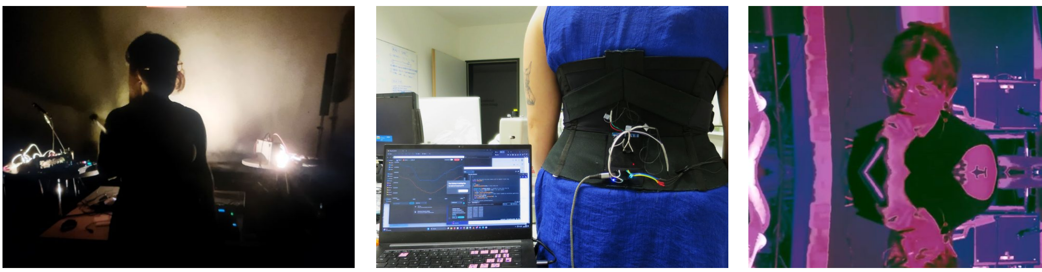

Fig. 1. Vocal BTEs constructed through movement sensing, my body, my classical vocal training, and electronic performance contexts. Photos thanks to Rachel Freire and Paul Strohmeier.

Vocal digital musical instruments (DMIs) have traditionally utilised movement and biosensing to capture performance gesture and technical practice in singing. However, the vocalist-voice relationship is entangled with social factors and pedagogy that shape individual practice. Likewise, technoscientific practices in adopting sensor technologies and creating action-to-sound mappings enforce particular conceptualisations of the body. Therefore, vocal DMIs are constructs of a variety of body conceptualisations and, as a result, shape interaction in different ways. Illustrated by vocal DMIs, this position argues that embodied interaction research must understand the construction of interactions before working to transform those experiences, and suggests paths to ensure body conceptualisations are included in design.

CCS Concepts: • **Human-centered computing → Interaction design theory, concepts and paradigms**; *HCI theory, concepts and models*; Interaction techniques.

Additional Key Words and Phrases: electromyography, agential realism, voice, sociotechnical, entanglement

ACM Reference Format:
Courtney N. Reed. 2026. (Re)Constructing the Body: Addressing Body Concepts in the Design of Vocal Technology. In *CHI 2026 Workshop on Body Transformation Experiences.* ACM, New York, NY, USA, 4 pages.

1 Introduction

In this position paper on Body Transformation Experiences (BTEs), introduce sociotechnical factors in experience. This expands on my contributions around Body Lutherie to the Body X Materials [8] and Sensorimotor Devices [16] workshop, which the current workshop builds on. I develop technology in vocal performance to shape user behaviour and conceptualisation of the musicking body; for instance, using sonified electromyography (EMG) signals to allow vocalists to "hear" their bodies and respond to their physiological movements in real-time [11, 13]. I have noted through this work that BTE research often focuses on experiences as a *result* of interaction. I counter this framing and demonstrate how BTEs are formed before technology is introduced into interaction paradigms. For instance, in the design and use of vocal DMIs, performance expectations and relationships with wearable technology inherently shape experiences and are entangled with BTE goals.

Body Transformation Experiences, Barcelona, Spain

To demonstrate the construction of experiences in vocal DMIs, I first discuss practices I have adopted in Human Computer Interaction (HCI) research for singing (Figure 1); namely, agential realism and the construction of experience through **Ambiguity Shifts** in design [10], and **Body Lutherie** practices to collaborate with the dynamic vocal body and its agency [5]. I demonstrate how values are encoded in vocal DMIs from artistic and HCI domains [9], meaning that BTEs are constructions of both existing conceptual framings and digital design and interactions. In doing so, I bring following questions around BTEs for discussion:

(1) How do designed factors (e.g., sensor mappings, multisensory data representations, etc.) meet pre-existing conceptualisations of the body and co-mingle in our design decisions?
(2) What are the values in these pre-existing domains and how can we aim to keep the desired aspects in BTEs without reinforcing negative biases and sociotechnical norms?

2 Experiences with Vocal Technology

Vocal DMIs have queried a variety of experiential factors in singing practice. Notably, Cotton et al. explored the "othering" of the body with movement-actuated wearables [6]; focusing on vocal breath practices, singer-body dynamics are shifted and can result in tension and estrangement [3]. Other vocal DMIs use non-vocal gestures; for instance, Imogen Heap's MiMu gloves [7] use hand gestures to manipulate the vocal signal. Other tactile and hand-held vocal controls—e.g., plucking [1] or mid-air interaction [17]—have also been used.

My previous work focused on externalisation of abstract, internal movements with EMG sonification [11, 12]. Autoethnographic perspectives of working with my own sonified movements revealed my internalised, subconscious gestures and shifted my performance behaviour [12]. However, there were mismatches between my conceptualisation as the system's designer and the conceptualisations of other vocalist users [13]. This revealed a plurality in embodied experiences and a pattern wherein vocalists, including myself, shifted behaviour towards interaction "encouraged" by the technology. In this way, the EMG *sensor* acted as an *actuator* [14].

Learning from these BTEs, I bring two points to exploring pre-existing body conceptualisations and how we can (1) manage Ambiguity Shifts in interaction, and (2) understand the convergence of factors in body-based interaction as Body Luthiers:

2.1 Ambiguity Shifts in BTEs

Taking inspiration from Barad's agential realism [2] and diffractive design practices [15], we can see how measuring a phenomena fixes it into a particular conceptualisation and collapses the multidimensional experience as we choose what to measure and how to map it. For instance, EMG, an ambiguous signal only loosely linked to a vocalist's conscious movement, becomes representative of a complex process. With another conceptualisation, this configuration would shift [10]. This is unavoidable to some degree (although we can endeavour to preserve ambiguity in particular places), so it is important to acknowledge how we *create* the experiences we claim to measure.

Unpacking these conceptualisations, it is important to note how my research is rooted in my own vocal practice and training; singers with different backgrounds will meet my conceptual mappings differently [13]. Sociocultural relations to our bodies—including movements, identity, shape, size, and so on [5]— and expectations of technology also inform interactions. We often expect technology to evaluate our "correctness", even when the goal of the BTE technology is for exploration and curiosity [13]. Likewise, the predominance of hand-based control in HCI has led to a dominance of tactile-based interactions, even when they are not otherwise relevant (e.g., in singing) [9, 17].

2.2 Body Lutherie Dynamics

Considering these sociocultural and technoscientific dimensions, my work has adopted a practice of Body Lutherie, a diffractive positioning of the physical body as a dynamic material in juxtaposition to the singer [5]. Taking inspiration from more-than-human design, this approach views the body as an entangled ecosystem. It endeavours to understand the multi-agential collaboration that must occur for vocal practice, and other movement-based or artistic practices, to emerge.

This approach was used in the design of a wearable corset, *Bones*, for my own vocal body [4, 5]. Bones captures diaphragmatic expansion across my abdomen and back, active in the supportive breathing necessary for singing. In addition to my experience, the corset was also developed according to other active agencies out of my control: the physical position of measurable muscle movement over time periods (e.g., with weight fluctuation, body shape changes), the day-to-day influence of both the external environment (e.g., stress, weather) and internal environment (e.g., microbiome, hormones), and the sociocultural expectations of my female body in performance. In this position, we as Body Luthiers explored the construction of experience and collaboration with, rather than control over, the body. Doing so, we honour the multifaceted agencies contributing to experience, even before the technology enters the picture.

3 Troubling the Technosolutionist

Utilising these two dimensions to *de*construct conceptualisations of the body, we can challenge technosolutionism in how we *re*construct the body in design and BTEs.

HCI often takes the technoscientific stance that, if we can optimise a sensor—its placement, mapping, etc.—we will incrementally get closer and closer to some kind of ground truth about experience [10]. However, body-based experiences are incredibly abstract, complex, and individual; to say that a sensor can measure and definitively capture an experience is reductive at best and physically, mentally, or emotionally harmful to a user at worst. We have a critical responsibility when we design BTEs to understand the way that technoscientific and sociocultural norms create our relationships with our bodies. In reality, sensor measurements and mappings provide a proxy for some aspects of experience, but never capture them fully. This kind of approach, however, can be incredibly helpful for exploration, play, and curiosity about the body and our experience, and allow us to trouble assumptions we make about those experiences.

To trouble the technosolutionist urge that can arise in HCI, I urge researchers and designers to consider the above approaches, which I have found success in, for the following actions:

First, we can examine fully how we construct interaction. By measuring the body and fixing its ambiguity, we inherently change our relationship with it. A bit like trying to hit a moving target, we constantly shift the interaction paradigm we are trying to measure. In this way, we need to consider how our sensors act as actuators, influencing the very behaviour we are trying to capture and becoming more and more entangled in BTEs.

Second, we need to meet the entangled social, biological, cultural, and technical factors halfway. Body-based interactions do not happen in a vacuum: we as designers must consider how we encourage views or create friction against conceptualisations. Productive frictions are necessary to challenge gendered roles and the reductive aspects of perceived identity that we carry with us in our bodies. Engaging BTE technology, we must be clear and informed about the other agencies at play in constructing experience. In taking a decisive role as researchers and designers on how we operate within these pre-existing constructions, we can design BTEs that align with our values and trouble assumptions for meaningful interaction.

References

[1] Maxwell Addae and Nina Masuelli. 2024. VocalCords: Exploring Tactile Interaction and Performance with the Singing Voice. In *Proceedings of the International Conference on New Interfaces for Musical Expression*, S M Astrid Bin and Courtney N. Reed (Eds.). Utrecht, Netherlands, Article 82, 8 pages. doi:10.5281/zenodo.13904951

[2] Karen Barad. 2007. *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning*. Duke University Press.

[3] Kelsey Cotton, Pedro Sanches, Vasiliki Tsaknaki, and Pavel Karpashevich. 2021. The Body Electric: A NIME designed through and with the somatic experience of singing. In *Proceedings of the International Conference on New Interfaces for Musical Expression*. Shanghai, China, Article 27. doi:10.21428/92fbeb44.ec9f8fdd

[4] Rachel Freire, Valentin Martinez-Missir, Courtney N. Reed, and Paul Strohmeier. 2024. RaveNET: Connecting People and Exploring Liminal Space through Wearable Networks in Music Performance. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork, Ireland) *(TEI '24)*. Association for Computing Machinery, New York, NY, USA, Article 89, 8 pages. doi:10.1145/3623509.3635270

[5] Rachel Freire and Courtney N. Reed. 2024. Body Lutherie: Co-Designing a Wearable for Vocal Performance with a Changing Body. In *Proceedings of the International Conference on New Interfaces for Musical Expression*, S M Astrid Bin and Courtney N. Reed (Eds.). Utrecht, The Netherlands, 117–126. doi:10.5281/zenodo.13904800

[6] Ozgun Kilic Afsar, Yoav Luft, Kelsey Cotton, Ekaterina R. Stepanova, Claudia Núñez Pacheco, Rebecca Kleinberger, Fehmi Ben Abdesslem, Hiroshi Ishii, and Kristina Höök. 2023. Corsetto: A Kinesthetic Garment for Designing, Composing for, and Experiencing an Intersubjective Haptic Voice. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 181, 23 pages. doi:10.1145/3544548.3581294

[7] Thomas Mitchell, Sebastian Madgwick, and Imogen Heap. 2012. Musical Interaction with Hand Posture and Orientation: A Toolbox of Gestural Control Mechanisms. In *Proceedings of the International Conference on New Interfaces for Musical Expression*. University of Michigan, Ann Arbor, Michigan. doi:10.5281/zenodo.1180543

[8] Courtney N. Reed. 2023. As the Luthiers Do: Designing with a Living, Growing, Changing Body-Material. In *ACM CHI Workshop on Body X Materials* (Hamburg, Germany).

[9] Courtney N. Reed. 2024. Enculturation and Value Encoding in the Design of Vocal DMIs. In *Proceedings of the CHIME Annual Conference (CHIME '24)*. The Open University, Milton Keynes, United Kingdom.

[10] Courtney N. Reed, Adan L. Benito, Franco Caspe, and Andrew P. McPherson. 2024. Shifting Ambiguity, Collapsing Indeterminacy: Designing with Data as Baradian Apparatus. *ACM Trans. Comput.-Hum. Interact.* 31, 6, Article 73 (Dec. 2024), 41 pages. doi:10.1145/3689043

[11] Courtney N. Reed and Andrew McPherson. 2020. Surface Electromyography for Direct Vocal Control. In *Proceedings of the International Conference on New Interfaces for Musical Expression*, Romain Michon and Franziska Schroeder (Eds.). Birmingham City University, Birmingham, UK, 458–463. doi:10.5281/zenodo.4813475

[12] Courtney N. Reed and Andrew P. McPherson. 2021. Surface Electromyography for Sensing Performance Intention and Musical Imagery in Vocalists. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Salzburg, Austria) *(TEI '21)*. Association for Computing Machinery, New York, NY, USA, Article 22, 11 pages. doi:10.1145/3430524.3440641

[13] Courtney N. Reed and Andrew P. McPherson. 2023. The Body as Sound: Unpacking Vocal Embodiment through Auditory Biofeedback. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction* (Warsaw, Poland) *(TEI '23)*. Association for Computing Machinery, New York, NY, USA, Article 7, 15 pages. doi:10.1145/3569009.3572738

[14] Courtney N. Reed, Landon Morrison, Andrew P. McPherson, David Fierro, and Atau Tanaka. 2024. Sonic Entanglements with Electromyography: Between Bodies, Signals, and Representations. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 2691–2707. doi:10.1145/3643834.3661572

[15] Pedro Sanches, Noura Howell, Vasiliki Tsaknaki, Tom Jenkins, and Karey Helms. 2022. Diffraction-in-action: Designerly Explorations of Agential Realism Through Lived Data. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 540, 18 pages. doi:10.1145/3491102.3502029

[16] Paul Strohmeier, Laia Turmo Vidal, Gabriela Vega, Courtney N. Reed, Alex Mazursky, Easa AliAbbasi, Ana Tajadura-Jiménez, and Jürgen Steimle. 2025. Sensorimotor Devices: Coupling Sensing and Actuation to Augment Bodily Experience. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 799, 7 pages. doi:10.1145/3706599.3706735

[17] Xiao Xiao and Sarah Fdili Alaoui. 2024. Tuning In to Intangibility: Reflections from My First 3 Years of Theremin Learning. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 2649–2659. doi:10.1145/3643834.3661584